

# Hidden Simplicity in Complex Patterns: Creating Fractals with The Collage Theorem

MATH 0723 Thesis

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## Abstract

Fractals are shapes that have endlessly repeating patterns. They typically have self-similarity, meaning that parts of the fractal are similar to the whole fractal, and they are often defined simply and recursively. Patterns that appear in nature like trees, clouds, and leaves also have self-similarity, and we can take advantage of their self-similarity to approximate them with simple, recursive functions. Barnsley's Collage Theorem is the tool we need to find such functions. In this paper, we build the mathematical theory that allows us to prove The Collage Theorem. We begin by learning about a convincing definition of the space of fractals. We prove that the space is complete and show how fractals can be built in this space with systems of contraction mappings called iterated function systems. Afterwards, we prove The Collage Theorem. Lastly, we see how The Collage Theorem is used for image compression. Throughout the paper, there are images and examples to help build intuition.

## Contents

|          |                                                      |           |
|----------|------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                  | <b>4</b>  |
| <b>2</b> | <b>The Metric Space of Fractals</b>                  | <b>7</b>  |
| 2.1      | Completeness of the Space of Fractals . . . . .      | 9         |
| <b>3</b> | <b>Contraction Mappings on the Space of Fractals</b> | <b>14</b> |
| <b>4</b> | <b>The Collage Theorem</b>                           | <b>20</b> |
| 4.1      | Continuous Changes in Attractors . . . . .           | 24        |
| <b>5</b> | <b>Fractal Image Compression</b>                     | <b>27</b> |
| <b>6</b> | <b>Conclusion</b>                                    | <b>30</b> |

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# 1 Introduction

Many shapes in nature are hard to represent with classical geometry. Trees, leaves, clouds, snowflakes, coastlines, and mountains seem almost random, but they can often be approximated with only a few functions. Fractals are good approximations of shapes in nature that have repeated patterns. The word *fractal* was first coined by the Polish-born mathematician Benoit B. Mandelbrot in his book *Les Objets Fractals* published in 1975 [1]. It is derived from the Latin word *fractus*, which means irregular or broken. Whereas much of math focuses on functions that are differentiable almost everywhere, fractals are abnormal because they have infinitely many kinks in their structure, making them non-differentiable on a set of non-zero measure. Fractals are hard to define, but they have some agreed upon properties [2, pp. xx-xxi].

1. Fractals have “fine structure,” meaning they have detail on infinitesimally small scales. You can zoom in any arbitrary amount on a fractal, and there will still be the same amount of detail.
2. Fractals are too irregular to be described with classical geometry either locally or globally.
3. Fractals exhibit some form of “self-similarity.” If you take a portion of the fractal, it is similar to the other parts of the fractal or the whole fractal.
4. Fractals can be thought of as having non-integer dimensions.
5. Fractals are often defined simply and perhaps recursively.

Let’s consider some well-known examples of fractals. A canonical example is the *Cantor set*, which exhibits all five properties above. Consider the following construction of the Cantor set [2, p. xiii], [3, p. 25]. Let  $E_0$  be the interval  $[0, 1]$  in  $\mathbb{R}$ , and let  $E_1$  be the set obtained by removing the middle third of the interval  $[0, 1]$ . That is,  $E_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$ . Continue repeating this pattern, so that  $E_k$  is obtained by removing the middle thirds of each interval in  $E_{k-1}$ . Then the Cantor set is defined as  $\bigcap_{k=1}^{\infty} E_k$ . The Cantor set exhibits fine structure (1) because it has detail at arbitrarily small scales. It is difficult to describe with classical geometry (2) but can be defined simply (5). The Cantor set has self-similarity (3); for example the portions of the set in  $[0, \frac{1}{3}]$  and  $[\frac{2}{3}, 1]$  are geometrically identical to the whole set but scaled by a factor of  $\frac{1}{3}$ . Lastly, classical descriptions of size like cardinality or measure can’t capture the Cantor set’s complexity because it is uncountable with cardinality  $2^{\aleph_0}$  (the same cardinality as  $[0, 1]$ ), yet it has measure zero. Thus, we need a new way to quantify size. Many mathematicians have done so by defining new fractal dimensions. The Cantor set’s dimension is intuitively something more complex than 0-dimensional but less complex than 1-dimensional. That is, we can think of the Cantor set as having a fractional dimension (4).

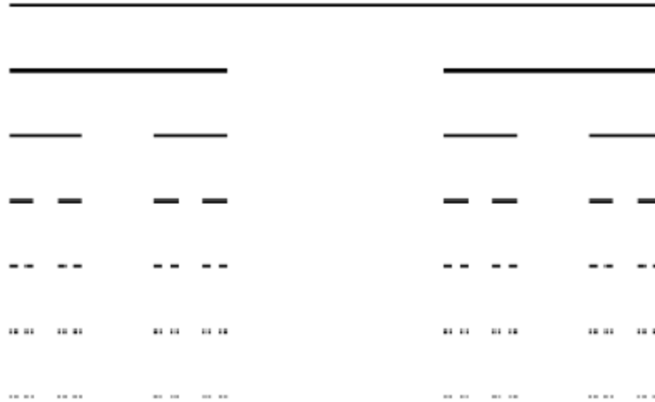


Figure 1: The first few sets in the construction of the Cantor set.  $E_0$  is at the top of the image, and  $E_6$  is at the bottom.

Another well-known example of a fractal is the *von Koch curve*, which also satisfies the five properties of a fractal. When constructing the von Koch curve, we also start with  $E_0 = [0, 1]$ . This time, we obtain  $E_1$  by removing the inner third of the interval and replacing it with two sides of an equilateral triangle. Then we construct  $E_2$  by applying the same process to each interval in  $E_1$ . The von Koch curve is the limit of this process as  $k$  goes to infinity.

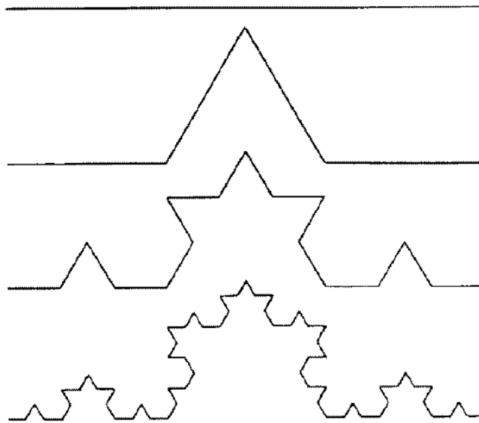


Figure 2: The first few sets in the construction of the von Koch curve where we start with  $E_0$  at the top and progress to  $E_3$  at the bottom.

The last fractal we consider for now is the *Sierpinski triangle*. To construct the Sierpinski triangle, we let  $E_0$  be a solid equilateral triangle in  $\mathbb{R}^2$ . We obtain  $E_1$  by

removing an upside down equilateral triangle from the center of  $E_0$ —we are left with three equilateral triangles. Similarly, we get  $E_2$  by removing an upside down equilateral triangle from the center of each triangle in  $E_1$ . The Sierpinski triangle is  $\bigcap_{k=1}^{\infty} E_k$ .

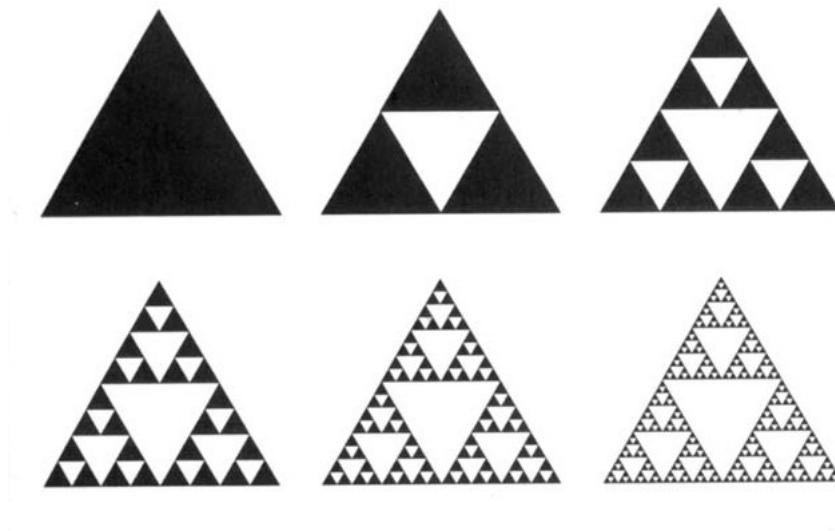


Figure 3: The first few sets in the construction of the Sierpinski Triangle where we start with  $E_0$  on the top left and continue to  $E_5$  on the bottom right.

It is intuitive that each of these fractals has fractional dimension, but how do we compute dimension? There are many definitions of fractal dimension, but some of the most common are the Hausdorff dimension, the box counting dimension, and the packing dimension [2]. The constructions of these definitions are outside the scope of this paper, but we can describe a simple notion of dimension called the *similarity dimension* [2, p. xix]. First, let's consider the Cantor set. It is comprised of two copies of itself that are each scaled by a factor of  $\frac{1}{3}$ . Therefore, the Cantor set has dimension  $-\log(2)/\log(\frac{1}{3})$ . Similarly, the von Koch curve is made of four copies of itself that are scaled by a factor of  $\frac{1}{3}$ , so it has dimension  $-\log(4)/\log(\frac{1}{3})$ , and the Sierpinski triangle is made of three copies of itself that are scaled by  $\frac{1}{2}$ ; therefore, it has dimension  $-\log(3)/\log(\frac{1}{2})$ . In general, if a fractal is constructed of  $k$  copies of itself that are each scaled by a factor of  $m$ , the fractal has similarity dimension  $-\log(k)/\log(m)$ .

Now, we have a good idea of what a fractal is. But as mathematicians we want a formal definition.... Unfortunately, there is no agreed upon definition of a fractal [2, p. xx]. Mandelbrot initially proposed that a fractal was any set whose Hausdorff dimension is greater than its topological dimension—which corresponds to Euclidean dimension in Euclidean space. This definition, however, excludes some sets that should clearly be considered fractals. Several other definitions have been proposed, but all of them seem to

have this same drawback.

To do math with fractals, we need to commit to one of the definitions. Section two defines the space of fractals as a metric space, and we use this definition throughout the paper. We also prove that the space of fractals is complete. Equipped with this knowledge, we show in section three that systems of contraction mappings—called *iterated function systems*—can be used to construct fractals. Section four explains how given any set, *The Collage Theorem* can be used to find a fractal approximation of the set. Then we see how continuous changes in contraction mappings create continuous changes in the fractals that they create. Lastly, section five describes how The Collage Theorem can be used for image compression.

## 2 The Metric Space of Fractals

There is no standard definition for the space of fractals. Any proposed definition seems to exclude shapes that should clearly be considered fractals. For our purposes, however, we use Barnsley’s definition of fractals in his book *Fractals Everywhere* [4].

**Definition 1.** *The Space of Fractals*

*Let  $(\mathbf{X}, d)$  be a complete metric space. The space of fractals  $\mathcal{H}(\mathbf{X})$  is the set of all compact subsets of  $\mathbf{X}$  other than the empty set.*

We can then use the metric  $d$  on  $\mathbf{X}$  to create a metric  $h(d)$  on  $\mathcal{H}(\mathbf{X})$ . We begin by defining the distance from a point to a set. Then we define the distance from a set to a set. Finally, we define the *Hausdorff distance*  $h(d)$  between two sets in  $\mathcal{H}(\mathbf{X})$ . Like any metric, the Hausdorff distance measures the similarity between two sets in  $\mathcal{H}(\mathbf{X})$  [4].

**Definition 2.** *Distance from a Point to a Set*

*Let  $(\mathbf{X}, d)$  be a complete metric space, and let  $x \in \mathbf{X}$  and  $B \in \mathcal{H}(\mathbf{X})$  be given. The distance from  $x$  to  $B$  is given by*

$$d(x, B) = \min\{d(x, y) : y \in B\}.$$

The minimum is guaranteed by the compactness of  $B$  [4].

**Definition 3.** *Distance from a Set to a Set*

*Let  $(\mathbf{X}, d)$  be a complete metric space, and let  $A, B \in \mathcal{H}(\mathbf{X})$  be given. The distance from  $A$  to  $B$  is given by*

$$d(A, B) = \max\{d(x, B) : x \in A\}.$$

Just as for Definition 2, the maximum of the points in  $A$  exists because  $A$  is compact [4]. We haven’t quite defined a metric yet because  $d(A, B)$  is not symmetric. That is, there exists sets  $A, B$  such that  $d(A, B) \neq d(B, A)$ . Accordingly, to make the Hausdorff metric symmetric, we define it in the following way.

**Definition 4.** *Hausdorff Distance*

The Hausdorff distance between  $A$  and  $B$  in  $\mathcal{H}(\mathbf{X})$  is defined by

$$h(A, B) = \max\{d(A, B), d(B, A)\}$$

**Proposition:** The Hausdorff distance is a metric on  $\mathcal{H}(\mathbf{X})$

**Proof that Hausdorff distance is a metric**

The following proof is adapted from Barnsley [4]. We need to show that Hausdorff distance satisfies the four properties that define a metric space. Let  $A, B, C \in \mathcal{H}(\mathbf{X})$  be given.

1.  $h(A, B) = \max\{d(A, B), d(B, A)\} = \max\{d(B, A), d(A, B)\} = h(B, A)$ . ✓
2. We want to show  $h(A, B) = 0 \iff A = B$   
 (  $\implies$  ) We prove this direction by proving the contrapositive: if  $A \neq B$ , then  $h(A, B) > 0$ . Suppose  $A \neq B$ . Then without loss of generality, there exists some  $a$  with  $a \in A$  and  $a \notin B$ . Thus  $h(A, B) \geq d(A, B) > 0$ .  
 (  $\impliedby$  ) If  $A = B$ ,  
 $h(A, B) = h(A, A) = \max\{d(A, A), d(A, A)\} = d(A, A) = \max\{d(x, A) : x \in A\} = 0$ .  
 ✓
3.  $h(A, B) \geq d(A, B) \geq 0$ . ✓
4. We want to show that  $h(A, C) \leq h(A, B) + h(B, C)$  for all  $A, B, C \in \mathcal{H}(\mathbf{X})$ . Let's begin by showing that  $d(A, B) \leq d(A, C) + d(B, C)$ . Fix  $a \in A$ . Then,

$$\begin{aligned} d(a, B) &= \min\{d(a, b) : b \in B\} \\ &\leq \min\{d(a, c) + d(c, b) : b \in B\} \forall c \in C \quad (\text{by the triangle inequality for } d). \\ &= [d(a, c) + \min\{d(c, b) : b \in B\}] \forall c \in C. \end{aligned}$$

Since the above inequality is true for all  $c \in C$ , it holds for the  $c$  that minimizes leftmost term,  $d(a, c)$ . Furthermore, taking the maximum of the rightmost term,  $\min\{d(c, b) : b \in B\}$ , only increases the right side of the inequality, so the inequality holds. We have that

$$\begin{aligned} d(a, B) &\leq \min\{d(a, c) : c \in C\} + \max\{\min\{d(c, b) : b \in B\} : c \in C\} \\ &= d(a, C) + d(C, B). \end{aligned}$$

Now, take the  $a$  that maximizes  $d(a, B)$  to get  $d(A, B)$ . Since  $d(a, C) \leq d(A, C)$  for all  $a$ ,

$$\begin{aligned} d(A, B) &\leq d(a, C) + d(C, B) \\ &\leq d(A, C) + d(C, B). \end{aligned}$$

We can finally show the triangle inequality for  $h$ .

$$\begin{aligned}
h(A, C) &= \max\{d(A, C), d(C, A)\} \\
&\leq \max\{d(A, B) + d(B, C), d(C, B) + d(B, A)\} \quad (\text{using the inequality above}) \\
&\leq \max\{d(A, B), d(B, A)\} + \max\{d(B, C), d(C, B)\} \\
&= h(A, B) + h(B, C). \checkmark
\end{aligned}$$

Since  $h$  satisfies 1-4 above,  $h$  is a metric. ■

## 2.1 Completeness of the Space of Fractals

In this section, our goal is to prove that the space of fractals is complete, meaning that all Cauchy sequences converge to a limit point  $A \in \mathcal{H}(\mathbf{X})$ .

**Theorem 2.1.** *The Completeness of the Space of Fractals*

*Let  $(\mathbf{X}, d)$  be a complete metric space. Then  $(\mathcal{H}(\mathbf{X}), h(d))$  is a complete metric space. Moreover, if  $\{A_n \in \mathcal{H}(\mathbf{X})\}_{n=1}^\infty$  is a Cauchy sequence, then*

$$A = \lim_{n \rightarrow \infty} A_n$$

*can be characterized by*

$$A = \{x \in \mathbf{X} : \text{there is a Cauchy sequence } \{x_n \in A_n\}_{n=1}^\infty \text{ that converges to } x\}.$$

Again, our proof is modified from Barnsley [4]. To prove theorem 2.1, we find a candidate limit point  $A$  for an arbitrary Cauchy sequence  $\{A_n\} \subset \mathcal{H}(\mathbf{X})$  and show both (1) that the sequence converges to the candidate  $A$  and (2) that the candidate  $A$  is in  $\mathcal{H}(\mathbf{X})$ . But first, we need a new definition, and a few lemmas.

**Definition 5.** *The Dilation of a Set*

*The dilation of a set  $L$  by a ball of radius  $\Gamma$  is given by*

$$S + \Gamma = \{y : d(x, y) \leq \Gamma \text{ for some } x \in S\}.$$

We can imagine that  $A + \Gamma$  is “stretching”  $A$  in every direction by a distance  $\Gamma$ . We have the following lemma about Hausdorff distance and dilations of sets. It not only helps us prove that  $\mathcal{H}(\mathbf{X})$  is complete, but it also improves our intuition about Hausdorff distance. It says that two sets  $A$  and  $B$  are distance  $\epsilon$  apart if and only if every point in  $A$  is less than or equal to distance  $\epsilon$  away from some point in  $B$ , and every point in  $B$  is less than or equal to distance  $\epsilon$  away from some point in  $A$ . The lemma is formally stated below.

**Lemma 2.1.1.** *Let  $A, B \in (\mathcal{H}(\mathbf{X}), h(d))$  be given where  $(X, d)$  is some metric space. Furthermore, let  $\epsilon > 0$  be given. Then*

$$h(A, B) \leq \epsilon \iff (A \subset B + \epsilon) \wedge (B \subset A + \epsilon)$$

*where  $\wedge$  denotes the logical “and”.*

### Proof of Lemma 2.1.2

Let  $A, B \in \mathcal{H}(\mathbf{X})$  be given where  $(X, d)$  is some metric space, and let  $\epsilon > 0$  be given. First, we show that  $d(A, B) \leq \epsilon \iff A \subset B + \epsilon$ .

$$\begin{aligned}
& d(A, B) \leq \epsilon \\
& \iff \max\{d(A, B), d(B, A)\} \leq \epsilon \\
& \iff (d(B, A) \leq \epsilon) \forall a \in A \\
& \iff A \subset B + \epsilon \quad \text{(by definition).}
\end{aligned}$$

By the same argument, we can show that  $d(B, A) \leq \epsilon \iff B \subset A + \epsilon$ . Finally, we have that

$$\begin{aligned}
& h(A, B) \leq \epsilon \\
& \iff \max\{d(A, B), d(B, A)\} \leq \epsilon \\
& \iff (d(A, B) \leq \epsilon) \wedge (d(B, A) \leq \epsilon) \\
& \iff (A \subset B + \epsilon) \wedge (B \subset A + \epsilon). \quad \square
\end{aligned}$$

The last lemma we need is known as *The Extension Lemma* and is discussed below.

**Lemma 2.1.2.** *Let  $(\mathbf{X}, d)$  be a metric space, and let  $\{A_n\}_{n=1}^\infty$  be a Cauchy sequence in  $(\mathcal{H}(\mathbf{X}), h(d))$ . Let  $\{n_j\}_{j=1}^\infty$  be a sequence of integers with  $n_1 < n_2 < n_3 \dots$ . Suppose we have a Cauchy sequence  $\{x_{n_j} \in A_{n_j}\}_{j=1}^\infty$  in  $(\mathbf{X}, d)$ . Then there exists a sequence  $\{\tilde{x}_n\}_{n=1}^\infty$  with each  $\tilde{x}_n \in A_n$  and  $\tilde{x}_{n_j} = x_{n_j}$ .*

### Proof

Let  $(\mathbf{X}, d)$  be a metric space, and let  $\{A_n\}_{n=1}^\infty$  be a Cauchy sequence in  $(\mathcal{H}(\mathbf{X}), h(d))$ . Let  $\{n_j\}_{j=1}^\infty$  be a sequence of integers with  $n_1 < n_2 < n_3 \dots$ . Suppose we have a Cauchy sequence  $\{x_{n_j}\}_{j=1}^\infty$  in  $(\mathbf{X}, d)$ . Let's construct  $\{\tilde{x}_n\}_{n=1}^\infty$  in the following way. For  $n = 1, 2, 3, \dots, n_1$ , let  $\tilde{x}_n$  be the point in  $A_n$  that is closest to  $x_{n_1}$  (i.e. the point that minimizes  $d(\tilde{x}_n, x_{n_1})$ ). The minimum is guaranteed to exist because  $A_n$  is compact. Repeat this process so that whenever  $n = n_{j-1} + 1, n_{j-1} + 2, \dots, n_j$ ,  $\tilde{x}_n$  is the point in  $A_n$  that minimizes  $d(\tilde{x}_n, x_{n_j})$ . Note that this distance is equal to  $d(x_{n_j}, A_n)$ . Now, we need to show that  $\{\tilde{x}_n\}$  satisfies the desired properties. By the sequence's construction we have that  $\tilde{x}_n \in A_n$  and  $\tilde{x}_{n_j} = x_{n_j}$ . Now, we need to show that  $\{\tilde{x}_n\}$  is Cauchy. Let  $\epsilon > 0$  be given. Recall that  $x_{n_j}$  is Cauchy by hypothesis, so there exists  $N_1 \in \mathbb{N}$  so that for all  $n_j, n_k \geq N_1$ ,  $d(x_{n_j}, x_{n_k}) < \epsilon/3$ . Similarly,  $\{A_n\}$  is Cauchy by hypothesis, so there exists  $N_2 \in \mathbb{N}$  so that  $h(A_n, A_m) < \epsilon/3$ . Take  $N = \max\{N_1, N_2\}$ . Now, notice that by the triangle inequality, we have that

$$d(\tilde{x}_n, \tilde{x}_m) \leq d(\tilde{x}_n, x_{n_j}) + d(x_{n_j}, x_{n_k}) + d(x_{n_k}, \tilde{x}_m).$$

From above, we have that  $d(x_{n_j}, x_{n_k}) < \epsilon/3$ . We also have that

$$\begin{aligned} d(\tilde{x}_n, x_{n_j}) &= d(x_{n_j}, A_n) \\ &\leq d(A_{n_j}, A_n) \\ &\leq h(A_{n_j}, A_n) \\ &< \epsilon/3. \end{aligned}$$

We can make the same argument to show that  $d(x_{n_k}, \tilde{x}_m) < \epsilon/3$ . In conclusion, we have that

$$\begin{aligned} d(\tilde{x}_n, \tilde{x}_m) &\leq d(\tilde{x}_n, x_{n_j}) + d(x_{n_j}, x_{n_k}) + d(x_{n_k}, \tilde{x}_m) \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon. \end{aligned}$$

Finally, we can prove that the space of fractals is complete

**Proof of Theorem 2.1**

Suppose that  $(X, d)$  is a complete metric space. Moreover, assume that  $\{A_n\}_{n=1}^\infty$  is a Cauchy sequence in  $(\mathcal{H}(\mathbf{X}), h(d))$ . Lastly, define  $A$ , our candidate for the limit point of  $\{A_n\}$ , as follows.

$$A = \{x : x \text{ is the limit point of a sequence } \{x_n \in A_n\}\}.$$

We prove the theorem by proving the following five facts.

1.  $A \neq \emptyset$ .
2.  $A$  is closed and therefore complete because  $\mathbf{X}$  is complete.
  - Recall that closed sets in complete metric spaces are complete [3, Th. 7.9].
3. for all  $\epsilon > 0$  there exists  $N \in \mathbb{N}$  such that for all  $n \geq N$ ,  $A \subset A_n + \epsilon$ .
4.  $A$  is totally bounded (and complete by (2)), so it is compact.
  - Recall that a set is totally bounded if and only if for all  $\epsilon > 0$ , we can find a finite set of open balls of radius epsilon whose union contains the set [3, p. 89]. Furthermore, recall that in a metric space, a set is compact if and only if it is totally bounded and complete [3, p. 108].
5.  $\lim_{n \rightarrow \infty} A_n = A$ .
  1. First, we show that that  $A$  is nonempty. Our goal is to find some Cauchy sequence  $\{x_n \in A_n\}$ . Because  $\mathbf{X}$  is complete, such a sequence converges to  $x$ , which is in  $A$

by our definition of  $A$ . By hypothesis,  $A_n$  is Cauchy, so we can find  $N_i \in \mathbb{N}$  so that for all  $m, n \geq N_i$ ,  $h(A_m, A_n) < \frac{1}{2^i}$ . This gives us a sequence of integers

$$N_1 < N_2 < \dots < N_k < \dots$$

so that  $h(A_{N_j}, A_{N_k}) < \frac{1}{2^i}$  for all  $j, k \geq i$ . By lemma 2.1.1  $A_{N_j} \subset A_{N_k} + \frac{1}{2^i}$  and  $A_{N_k} \subset A_{N_j} + \frac{1}{2^i}$ , so for all  $x_{N_j} \in A_{N_j}$  and  $x_{N_k} \in A_{N_k}$ ,  $d(x_{N_j}, x_{N_k}) \leq \frac{1}{2^i}$ . Therefore, we can construct a sequence by taking an arbitrary  $x_{N_i} \in A_{N_i}$ . We know that such an  $x_{N_i}$  exists because  $A_{N_i} \in \mathcal{H}(\mathbf{X})$ , and  $\mathcal{H}(\mathbf{X})$  does not contain the empty set by definition. Moreover,  $x_{N_i}$  is Cauchy since for every  $\epsilon > 0$ , we can choose  $i$  large enough so that  $\frac{1}{2^i} < \epsilon$  and therefore  $d(x_{N_j}, x_{N_k}) < \epsilon$  for every  $j, k \geq i$ . Since  $\mathbf{X}$  is complete by hypothesis,  $\{x_{N_i}\}$  converges to some  $x \in \mathbf{X}$ . By lemma 2.1.2, we can construct a sequence  $\{\tilde{x}_n\}$  such that each  $\tilde{x}_n$  is in  $A_n$ ,  $\tilde{x}_{N_i} = x_{N_i}$ , and  $\{\tilde{x}_n\}$  converges to  $x$ . By definition of  $A$ ,  $x \in A$ , so  $A$  is nonempty.  $\checkmark$

2. Second, we want to show that  $A$  is closed. Let  $\{a_n\} \subset A$  be a sequence that converges to some point  $a \in \mathbf{X}$ . Since  $\{a_n\}$  is arbitrary, if we show that  $a \in A$ , we have shown that all convergent sequences in  $A$  have limit points in  $A$ , meaning that  $A$  is closed by definition. Since each  $a_n$  is in  $A$ , by our definition of  $A$  there exists a sequence  $\{a_{i,k}\}_{i=1}^\infty$  with each  $a_{i,k} \in A_i$  such that  $\{a_{i,k}\}$  converges to an element  $a_k$  in the sequence  $\{a_n\}$ . Because  $\{a_n\}$  converges, there exists a strictly increasing sequence of integers  $\{N_i\}$  so that  $d(a_{N_i}, a) < 1/i$ . And since  $\{a_{i,k}\}$  converges, there exists a corresponding increasing sequence of integers  $\{m_i\}$  so that  $d(a_{m_i, N_i}, a_{N_i}) < 1/i$  for every  $N_i$ . Therefore, by the triangle inequality,  $d(a_{m_i, N_i}, a) \leq d(a_{m_i, N_i}, a_{N_i}) + d(a_{N_i}, a) < 2/i$ . Let  $\{y_i\}$  be a sequence defined by  $\{y_i = a_{N_i, m_i}\}_{i=1}^\infty$ . Then each  $y_i \in A_{m_i}$ , and the sequence converges to  $a$ . By lemma 2.1.2, there exists a sequence  $\{\tilde{y}_n \in A_n\}$  that converges to  $a$ ; therefore,  $a \in A$  by definition.  $\checkmark$
3. Now, we show that for all  $\epsilon > 0$  there exists  $N \in \mathbb{N}$  so that for all  $n \geq N$   $A \subset A_n + \epsilon$ . Let  $\epsilon > 0$  be given. Since  $\{A_n\}$  is Cauchy, there exists some  $N$  such that for all  $m > n \geq N$ ,  $h(A_m, A_n) < \epsilon$ . By lemma 2.1.1, this implies that  $A_m \subset A_n + \epsilon$ . Let  $a \in A$  be given. Our goal is to show that  $a \in A_n + \epsilon$ . Since  $a \in A$  there exists some sequence  $\{a_n \in A_n\}$  so that  $a_n$  converges to  $a$ . Suppose our  $N$  is also sufficiently large enough so that for all  $m > n \geq N$ ,  $d(a_m, a) < \epsilon$ . Therefore the sequence starting at  $m > n$ ,  $\{a_m\}$  is a subset  $A_n + \epsilon$ . Now, notice that because  $A_n$  is compact, it is closed. Thus,  $A_n + \epsilon$  is also closed, so  $a$ , the limit point of  $\{a_m\}$ , is contained in  $A$ . Since  $a$  was arbitrary,  $A \subset A_n + \epsilon$ .  $\checkmark$
4. We want to show that  $A$  is totally bounded. Since totally bounded and complete sets are compact, combining this proof with (2), we get that  $A$  is compact. We prove that  $A$  is totally bounded by contradiction. Suppose that  $A$  is not totally

bounded. Then for some  $\epsilon > 0$ , no finite set of  $\epsilon$ -balls can cover  $A$ . Therefore, we can find a sequence  $\{x_n\} \subset A$  with  $d(x_i, x_j) \geq \epsilon$  for all  $i, j \in \mathbb{N}$  with  $i \neq j$ , telling us that no two points in this sequence can be contained in the same  $\epsilon$ -ball. By (3), there exists  $N \in \mathbb{N}$  such that for all  $n \geq N$ ,  $A \subset A_n + \epsilon/3$ . Fix such an  $n \geq N$ . Then for all  $x_i \in \{x_n\}$ , there exists some  $y_i \in A_n$  with  $d(x_i, y_i) < \epsilon/3$ . Since  $A_n$  is compact, the sequence  $\{y_i\}$  has a convergent subsequence  $\{y_{n_i}\}$ . Thus, we can find two points  $y_{n_k}$  and  $y_{n_j}$  as close as we would like. In particular, we can find points so that  $d(y_{n_k}, y_{n_j}) < \epsilon/3$ . But then we have

$$\begin{aligned} d(x_{n_k}, x_{n_j}) &\leq d(x_{n_k}, y_{n_k}) + d(y_{n_k}, y_{n_j}) + d(y_{n_j}, x_{n_j}) \\ &< \epsilon/3 + \epsilon/3 + \epsilon/3 = \epsilon, \end{aligned}$$

which is a contradiction.  $\checkmark$

5. Lastly, we want to show that  $\lim_{n \rightarrow \infty} A_n$  by proving that for all  $\epsilon > 0$  there exists  $N \in \mathbb{N}$  so that  $d(A_n, A) \leq \epsilon$ . We can use less than or equal to instead of the standard strict inequality because it is true for all epsilon. By lemma 2.1.1,  $d(A_n, A) \leq \epsilon$  if and only if  $A \subset A_n + \epsilon$  and  $A_n \subset A + \epsilon$ . We have already shown that  $A \subset A_n + \epsilon$  in part (3), so we just need to show that  $A_n \subset A + \epsilon$ . Because  $\{A_n\}$  is Cauchy, there exists some  $N \in \mathbb{N}$  so that for all  $n \geq N$ ,  $h(A_n, A_m) < \epsilon/4$ . Fix such an  $n \geq N$ , and let  $y \in A_n$  be given. There is also a sequence of integers  $\{N_i\}$  with  $n < N_1 < N_2 < \dots$  such that for all  $k > s \geq N_j$ ,  $h(A_k, A_s) < \frac{\epsilon}{2^{j+2}}$ . By lemma 2.1.1, we have that  $A_n \subset A_{N_1} + \epsilon/4$  by our choice of  $n$ , and  $A_s \subset A_k + \frac{\epsilon}{2^{j+2}}$ . Therefore, there exists some  $x_{N_1} \in A_{N_1}$  with  $d(x_{N_1}, y) \leq \epsilon/4$ . Similarly, for every  $j \in \mathbb{N}$  there exists some  $x_{j+1} \in A_{j+1}$  so that  $d(x_j, x_{j+1}) \leq \frac{\epsilon}{2^{j+2}}$ . By induction, we can construct a sequence of such  $x_{N_j}$ 's. Notice that the distance  $d(x_{N_j}, y)$  for any point in the sequence  $x_{N_j}$  is given by

$$\begin{aligned} d(x_{N_j}, y) &\leq d(x_{N_1}, y) + \sum_{k=1}^{j-1} d(x_{N_k}, x_{N_{k+1}}) && \text{(by the triangle inequality)} \\ &< \epsilon/4 + \sum_{k=1}^{\infty} \frac{\epsilon}{2^{k+2}} \\ &= \epsilon/4 + \epsilon/4 && \text{(sum of a geometric series)} \\ &= \epsilon/2 \end{aligned}$$

Moreover,  $\{x_{N_j}\}$  is Cauchy, since the distance between any two points  $x_{N_i}$  and

$x_{N_{t+c}}$  for some  $c \in \mathbb{N}$  is characterized by

$$\begin{aligned}
d(x_t, x_{t+c}) &\leq \sum_{k=t}^{t+c-1} d(x_{N_k}, d_{N_{k+1}}) && \text{(by the triangle inequality)} \\
&= \sum_{k=t}^{t+c-1} \frac{\epsilon}{2^{k+2}} \\
&< \sum_{k=t}^{\infty} \frac{\epsilon}{2^{k+2}} \\
&= \frac{\epsilon}{2^{t+1}} && \text{(sum of a geometric series),}
\end{aligned}$$

so we can choose  $t$  large enough to make this distance arbitrarily small. Since  $\mathbf{X}$  is complete,  $\{x_{N_j}\}$  converges to some element  $x \in \mathbf{X}$ , and we can choose  $j$  sufficiently large enough, so  $d(x_{N_j}, x) < \epsilon/2$ . Additionally,  $x \in A$  since by lemma 2.1.2 we can create a corresponding sequence  $\{\tilde{x}_n\}$  that converges to  $x$  with each element  $\tilde{x}_n \in A_n$ . Now, by the triangle inequality, we have that

$$d(x, y) \leq d(x, x_{N_j}) + d(x_{N_j}, y) < \epsilon/2 + \epsilon/2 = \epsilon.$$

Since we found some  $x \in A$  with  $d(x, y) < \epsilon$ ,  $y \in A + \epsilon$ , and since our choice of  $y \in A_n$  was arbitrary,  $A_n \subset A + \epsilon$ . ■

### 3 Contraction Mappings on the Space of Fractals

We have shown that the space of fractals with the Hausdorff metric is complete, so by definition, any Cauchy sequence in  $\mathcal{H}(\mathbf{X})$  converges to some limit point  $A \in \mathcal{H}(\mathbf{X})$ . We use this fact to construct new fractals, which are the limit points of sequences. In particular, we construct sequences by iteratively taking the image of a set under some function  $w$ . That is, our sequence is given by  $w^n(A)$  for some set  $A \in \mathcal{H}(\mathbf{X})$  where  $w^n$  denotes the composition of  $w$  with itself  $n$  times. We don't, however, want our sequence to "blow up" as  $n$  gets large, so we restrict our functions to be *contraction mappings*. Contraction mappings are a special type of function that shrinks the distance between any two points in a set. In Theorem 3.2 (*The Contraction Mapping Theorem*) we prove that if  $w$  is a contraction mapping then  $w^n(A)$  converges for any  $A \in \mathcal{H}(\mathbf{X})$ . Fractals that are limits of these sequences are called *deterministic fractals*.

**Definition 6.** *Contraction Mapping*

Let  $(\mathbf{X}, d)$  be a metric space. A contraction mapping  $w : \mathbf{X} \rightarrow \mathbf{X}$  is a function such that

$$d(w(x), w(y)) \leq s \cdot d(x, y)$$

for all  $x, y \in \mathbf{X}$  and some constant  $0 \leq s < 1$ . The constant  $s$  is called a contractivity factor for  $w$ .

Intuitively, contraction mappings shrink the distance between any two points. If we take the image of a set under a contraction mapping, every pair of points in the image is closer together than they were before the transformation. Contraction mappings are also continuous, which will be important for our proof of Theorem 3.2 (*The Contraction Mapping Theorem*).

**Theorem 3.1.** *The Continuity of Contraction Mappings*

Let  $w : \mathbf{X} \rightarrow \mathbf{X}$  be a contraction mapping on the metric space  $(X, d)$ . Then  $w$  is continuous.

**Proof of Theorem 3.1**

Let  $(\mathbf{X}, d)$  be a metric space, let  $x, y \in \mathbf{X}$  be given, and let  $w : (\mathbf{X}, d) \rightarrow (\mathbf{X}, d)$  be a contraction mapping with contractivity factor  $s$ . Let  $\epsilon > 0$  be given and take  $\delta = \epsilon/s$ . Whenever  $d(x, y) < \delta = \epsilon/s$ , we have that  $s \cdot d(x, y) < \epsilon$ . By the definition of a contraction mapping,

$$d(w(x), w(y)) \leq s \cdot d(x, y) < \epsilon.$$

Therefore,  $w$  is continuous by definition. ■

**Theorem 3.2.** *The Contraction Mapping Theorem*

Let  $w : \mathbf{X} \rightarrow \mathbf{X}$  be a contraction mapping on the complete metric space  $(\mathbf{X}, d)$ . Then  $w$  has exactly one fixed point  $x'$ , and for any  $x \in \mathbf{X}$ ,

$$\lim_{n \rightarrow \infty} w^n(x) = x'.$$

Here  $w^n$  denotes the composition of  $w$  with itself  $n$  times.

**Proof of Theorem 3.2**

Our proof is modified from Barnsley [4]. Let  $x \in \mathbf{X}$  be given and suppose  $w : \mathbf{X} \rightarrow \mathbf{X}$  is a contraction mapping with contractivity factor  $s$ . First, we show that  $w^n(x)$  is a convergent sequence of functions. Let  $m, n \in \mathbb{N}$  be given, and suppose that  $n < m$  (without loss of generality).

$$d(w^m(x), w^n(x)) = d(w^n(w^{m-n}(x)), w^n(x)) \tag{1}$$

$$\leq s^n \cdot d(w^{m-n}(x), x). \tag{2}$$

Now, notice that for all  $k \in \mathbb{N}$

$$d(x, w^k(x)) \tag{3}$$

$$\leq d(x, w(x)) + d(w(x), w^2(x)) + d(w^2(x), w^3(x)) + \dots + d(w^{k-1}(x), w^k(x)) \tag{4}$$

$$\leq (1 + s + s^2 + s^3 + \dots + s^{k-1})d(x, w(x)) \tag{5}$$

$$< (1 - s)^{-1} \cdot d(x, w(x)) \tag{6}$$

where we get from (3) to (4) by the triangle inequality for  $d$ . We get from (4) to (5) using the definition of a contraction mapping and the fact that  $w^k(x) = w^{k-1} \circ w(x)$ . And we get from (5) to (6) because an infinite geometric series  $1 + s + s^2 + \dots$  converges to  $(1 - s)^{-1}$ , and we are only taking the first  $k$  terms. Substituting the above inequality into (1)(2), we get that

$$d(w^m(x), w^n(x)) \leq \frac{s^n}{1 - s} \cdot d(x, w(x)). \quad (7)$$

Let  $\epsilon > 0$  be given. Since  $\frac{1}{1-s} \cdot d(x, w(x))$  is constant and  $0 \leq s < 1$ , we can choose  $N \in \mathbb{N}$  large enough so that for all  $m, n \geq N$

$$d(w^m(x), w^n(x)) \leq \frac{s^n}{1 - s} \cdot d(x, w(x)) < \epsilon.$$

Therefore,  $w^n(x)$  is Cauchy by definition. Since  $X$  is complete,  $w^n$  also converges to some limit  $x' \in \mathbf{X}$ , so we have

$$\lim_{n \rightarrow \infty} w^n(x) = x'.$$

Now, we show that  $x'$  is a fixed point. Since contraction mappings are continuous, the following equalities are valid:

$$w(x') = w\left(\lim_{n \rightarrow \infty} w^n(x)\right) = \lim_{n \rightarrow \infty} w(w^n(x)) = \lim_{n \rightarrow \infty} w^{n+1}(x) = x'.$$

Lastly, we must show that  $x'$  is the only fixed point. Suppose that  $x'$  and  $y'$  are both fixed points (i.e.  $w(x') = x'$  and  $w(y') = y'$ ). Then

$$d(x', y') = d(w(x'), w(y')) \leq s \cdot d(x', y'). \quad (8)$$

This implies that  $(1 - s)d(x', y') \leq 0$ . Since  $0 \leq s < 1$ ,  $(1 - s) > 0$ ; therefore,  $d(x', y') = 0$ . This is only possible if  $x' = y'$  because  $d$  is a metric. ■

Now, we know that contraction mappings have some important properties. Naturally, contraction mappings on  $(\mathbb{R}, |\cdot|)$  where  $|\cdot|$  is Euclidean distance are among the easiest to construct whereas Hausdorff distance is harder to intuit and work with. Fortunately, we can construct a contraction mapping on  $(\mathcal{H}(\mathbf{X}), h(d))$  from contraction mappings on  $(\mathbb{R}, |\cdot|)$  in the following way [4].

**Theorem 3.3.** *Constructing Contraction Mappings on the Space of Fractals*

*Let  $w : \mathbf{X} \rightarrow \mathbf{X}$  be a contraction mapping on the metric space  $(\mathbf{X}, d)$  with contractivity factor  $s$ . Then  $w : \mathcal{H}(\mathbf{X}) \rightarrow \mathcal{H}(\mathbf{X})$  defined by*

$$w(B) = \{w(x) : x \in B\}$$

*for all  $B \in \mathcal{H}(\mathbf{X})$  is a contraction mapping on the metric space  $(\mathcal{H}(\mathbf{X}), h(d))$  with contractivity factor  $s$ . Notice that  $w(B)$  is just the image of  $B$ .*

### Proof of Theorem 3.3

Let  $w : \mathbf{X} \rightarrow \mathbf{X}$  be a contraction mapping on the metric space  $(\mathbf{X}, d)$  with contractivity factor  $s$ . First, notice that our function maps to the correct codomain;  $w$  is continuous, so the image of any compact set is compact. Now, we want to show that  $h(w(A), w(B)) \leq s \cdot h(A, B)$ . Let's begin by showing that

$$d(w(A), w(B)) \leq s \cdot d(A, B).$$

Notice that

$$d(w(A), w(B)) = \max\{\min\{d(w(x), w(y)) : w(y) \in w(B)\} : w(x) \in w(A)\} \quad (1)$$

$$= \max\{\min\{d(w(x), w(y)) : y \in B\} : x \in A\} \quad (2)$$

$$\leq \max\{\min\{s \cdot d(x, y) : y \in B\} : x \in A\} \quad (3)$$

$$= s \cdot \max\{\min\{d(x, y) : y \in B\} : x \in A\} \quad (4)$$

$$= s \cdot d(A, B). \quad (5)$$

We use the fact that  $w$  is a contraction mapping moving from (2) to (3). Similarly, we can show that  $d(w(B), w(A)) \leq s \cdot d(B, A)$ . Using this information, we have that

$$\begin{aligned} h(w(A), w(B)) &= \max\{d(w(A), w(B)), d(w(B), w(A))\} \\ &\leq \max\{s \cdot d(A, B), s \cdot d(B, A)\} \\ &= s \cdot \max\{d(A, B), d(B, A)\} \\ &= s \cdot h(A, B). \end{aligned} \quad \blacksquare$$

The next theorem from Barnsley [4] shows us that we can combine contraction mappings on  $\mathcal{H}(\mathbf{X})$  to create new ones.

### Theorem 3.4. Constructing New Contraction Mappings

Let  $(\mathbf{X}, d)$  be a metric space and let  $w_1, w_2, \dots, w_n$  be a finite number of contraction mappings on  $(\mathcal{H}(\mathbf{X}), h(d))$ . Let the contractivity factor for each  $w_k$  be denoted with  $s_k$ . Define  $W : \mathcal{H}(\mathbf{X}) \rightarrow \mathcal{H}(\mathbf{X})$  with

$$\begin{aligned} W(B) &= w_1(B) \cup w_2(B) \cup \dots \cup w_n(B) \\ &= \bigcup_{i=1}^n w_i(B). \end{aligned}$$

Then  $W$  is a contraction mapping with contractivity factor  $s = \max\{s_1, s_2, \dots, s_n\}$ .

To prove this theorem, we need a few lemmas.

**Lemma 3.4.1.** Let  $A$ ,  $B$ , and  $C$  be subsets of some metric space  $(\mathbf{X}, d)$ . Then

$$d(A \cup B, C) = \max\{d(A, C), d(B, C)\}. \quad (6)$$

**Proof of Lemma 3.4.1**

Let  $(\mathbf{X}, d)$  be a metric space, and let  $A, B, C \subset \mathbf{X}$  be given. We have that

$$\begin{aligned} d(A \cup B, C) &= \max\{d(x, C) : x \in A \cup B\} && \text{(by definition)} \\ &= \max\{\{d(x, C) : x \in A\} \cup \{d(x, C) : x \in B\}\} \\ &= \max\{d(A, C), d(B, C)\}. \end{aligned} \quad \square$$

**Lemma 3.4.2.** *Let  $(\mathbf{X}, d)$  be a metric space, and let  $A, B, C$ , and  $D$  be sets in the metric space  $(\mathcal{H}(\mathbf{X}), h(d))$ . Then*

$$h(A \cup B, C \cup D) \leq \max\{h(A, C), h(B, D)\}$$

**Proof of Lemma 3.4.2**

Let  $(\mathbf{X}, d)$  be a metric space, and let  $A, B, C, D \in \mathcal{H}(\mathbf{X})$  be given. First, we prove the inequality that  $d(A \cup B, C \cup D) \leq \max\{d(A, C), d(B, D)\}$ . The desired inequality easily follows. From Lemma 3.4.1, we have that

$$d(A \cup B, C \cup D) = \max\{d(A, C \cup D), d(B, C \cup D)\}.$$

Now, let's consider the terms inside the maximum function on the right hand side.

$$\begin{aligned} d(A, C \cup D) &= \max\{\min\{d(x, y) : y \in C \cup D\} : x \in A\} \\ &\leq \max\{\min\{d(x, y) : y \in C\} : x \in A\} \\ &= d(A, C). \end{aligned}$$

Similarly,  $d(B, C \cup D) \leq d(B, D)$ . Therefore, we have that  $d(A \cup B, C \cup D) \leq \max\{d(A, C), d(B, D)\}$ . Now consider the following.

$$\begin{aligned} h(A \cup B, C \cup D) &= \max\{d(A \cup B, C \cup D), d(C \cup D, A \cup B)\} \\ &\leq \max\{\max\{d(A, C), d(B, D)\}, \max\{d(C, A), d(D, B)\}\} \\ &= \max\{d(A, C), d(B, D), d(C, A), d(D, B)\} \\ &= \max\{\max\{d(A, C), d(C, A)\}, \max\{d(B, D), d(D, B)\}\} \\ &= \max\{h(A, C), d(B, D)\}. \end{aligned} \quad \square$$

Now, we can prove theorem 3.4.

**Proof of Theorem 3.4**

Let  $(\mathbf{X}, d)$  be a metric space, and let  $w_1, w_2, \dots, w_n$  be contraction mappings with the contractivity factor for each  $w_k$  denoted  $s_k$ . Define  $W : \mathcal{H}(\mathbf{X}) \rightarrow \mathcal{H}(\mathbf{X})$  by  $W(B) = \bigcup_{i=1}^n w_i(B)$ . Notice that the union of compact sets is compact, so  $W$  has the correct codomain. We can prove the theorem using induction on the number of contraction mappings  $n$ . For our base case, let  $n = 1$ . Then  $W(B) = w_1(B)$ ; since  $w_1(B)$  is a contraction mapping with contractivity factor  $s_1$  (by hypothesis),  $W(B)$  is also a

contraction mapping with contractivity factor  $s = \max\{s_1\} = s_1$ . Take the following induction hypothesis: suppose that  $W_k$  is a contraction mapping for some  $k \in \mathbb{N}$  with contractivity factor  $s'_k = \max\{s_1, s_2, \dots, s_k\}$ . Then

$$W_{k+1} = W_k(B) \cup w_{k+1}(B).$$

We want to show that  $W_{k+1}$  is also a contraction mapping with contractivity factor  $s = \max\{s_1, s_2, \dots, s_{k+1}\}$ . Let  $A, B \in \mathcal{H}(\mathbf{X})$  be given. Then

$$\begin{aligned} & h(W_k(B) \cup w_{k+1}(B), W_k(A) \cup w_{k+1}(A)) \\ & \leq \max\{h(W_k(B), W_k(A)), h(w_{k+1}(B), w_{k+1}(A))\} \quad (\text{by Lemma 3.4.2}) \\ & \leq \max\{s'_k \cdot h(B, A), s_{k+1} \cdot h(B, A)\} \\ & = \max\{s'_k, s_{k+1}\} \cdot h(B, A) \end{aligned}$$

where  $\max\{s'_k, s_{k+1}\} = \max\{s_1, s_2, \dots, s_k, s_{k+1}\} = s$ . In conclusion, we have shown that for all  $n$ ,  $W(B) = \bigcup_{i=1}^n w_i(B)$  is a contraction mapping with contractivity factor  $s = \max\{s_1, s_2, \dots, s_n\}$ . ■

For any finite contraction mappings  $w_1, w_2, \dots, w_n$  on the complete metric space  $(\mathbf{X}, d)$ , since  $W(B) = \bigcup_{i=1}^n w_i(B)$  is a contraction mapping, the contraction mapping theorem applies.  $W$  has exactly one fixed point  $A$ , and for all  $B \in \mathcal{H}(\mathbf{X})$ ,  $\lim_{n \rightarrow \infty} W^n(B) = A$ . Since any such set of contraction mappings has these properties, we give it a special name: an *iterated function system (IFS)*.

**Definition 7.** *Iterated Function System (IFS)*

An iterated function system (IFS) is a complete metric space  $(\mathbf{X}, d)$  together with a finite set of contraction mappings  $w_1, w_2, \dots, w_n$  on  $\mathbf{X}$ .

We called the fixed point of an IFS the *attractor* of the IFS [4].

Let's consider an example. What's the attractor of the IFS consisting of the contraction mappings  $w_1, w_2, w_3$  on  $\mathbb{R}^2$  below?

$$\begin{aligned} w_1(x, y) &= \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\ w_2(x, y) &= \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ 1/2 \end{bmatrix} \\ w_3(x, y) &= \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}. \end{aligned}$$

Imagine we start with the compact set  $A = [0, 1] \times [0, 1] \subset \mathbb{R}^2$ . Let  $W(A) = \bigcup_{i=1}^3 w_i(A)$ . Notice that  $w_1$  simply shrinks the square to  $[0, 1/2] \times [0, 1/2]$ ;  $w_2$  shrinks the square and shifts it up by  $1/2$ ; and  $w_3$  shrinks the square and shifts it both upwards by  $1/2$  and to the right by  $1/2$ . When we repeat this process,  $W^2(A)$  shrinks the entire shape in an

analogous way. We can repeat this process and the limit  $\lim_{n \rightarrow \infty} W^n(A)$  converges to the shape depicted in Figure 4, the Sierpinski Triangle!

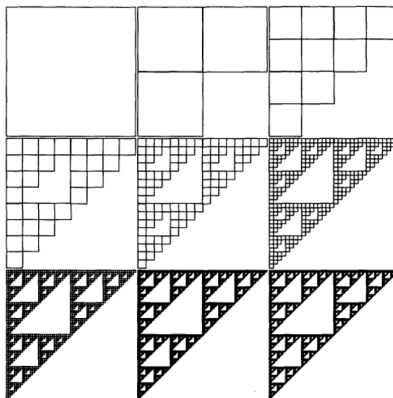


Figure 4: The IFS described above represents the Sierpinski Triangle with just three simple functions [4].

Note, that we could have set  $A$  to be *any* compact set. We could have started with  $A$  as a circle, triangle, rectangle, star, or any closed and bounded blob in  $\mathbb{R}^2$ , and  $\lim_{n \rightarrow \infty} W^n(A)$  would have converged to the Sierpinski Triangle because it is the sole attractor of the IFS.

## 4 The Collage Theorem

We can now construct fractals with an IFS, but if we have a fractal that we want to represent with an IFS, it's hard to imagine how. Additionally, we may want to find a fractal that approximate non-fractal shapes. For example, how might we approximate the leaf depicted in Figure 5?



Figure 5: How could we represent this leaf with a fractal?

The *Collage Theorem* [4] gives us an easy way to find the IFS of a fractal. It says that if we find contraction mappings on a set whose union is close to the original set, then the attractor of the contraction mappings is also close to the original set. In this section, we begin by stating and proving the theorem. Then we look at examples that show how we can use the theorem.

**Theorem 4.1.** *The Collage Theorem*

Let  $L \in \mathcal{H}(\mathbf{X})$ , and let  $\epsilon \geq 0$  be given. Choose an IFS  $w_1, w_2, \dots, w_n$  with attractor  $A$  and contractivity factor  $s$  so that

$$h(L, \bigcup_{i=1}^n w_i(L)) \leq \epsilon.$$

Then

$$h(L, A) \leq \epsilon/(1 - s).$$

Practically, we have some set  $L$  that we want to approximate with a deterministic fractal. If the union of several contraction mappings on  $L$  is close to  $L$ , then the attractor of the contraction mappings is also close to  $L$ . We follow Barnsley's proof of The Collage Theorem [4], which uses the following lemma.

**Lemma 4.1.1.** *Let  $(\mathbf{X}, d)$  be a complete metric space, and let  $w : \mathbf{X} \rightarrow \mathbf{X}$  be a contraction mapping with contractivity factor  $s$ . Furthermore, suppose  $x'$  is the fixed point of  $w$ . Then for all  $x \in \mathbf{X}$ ,*

$$d(x, x') \leq (1 - s)^{-1} \cdot d(x, w(x)).$$

**Proof of Lemma 4.1.1**

Let  $w$  be a contraction mapping on  $(\mathbf{X}, d)$  with fixed point  $x'$  and contractivity factor  $s$ . Then

$$\begin{aligned} d(x, x') &= d(x, \lim_{n \rightarrow \infty} w^n(x)) \\ &\leq d(x, w(x)) + \sum_{k=1}^{\infty} d(w^k(x), w^{k+1}(x)) \quad (\text{by the triangle inequality}) \\ &\leq \sum_{k=0}^{\infty} s^k \cdot d(x, w(x)) \quad (\text{using the definition of a contraction mapping}) \\ &= (1 - s)^{-1} \cdot d(x, w(x)) \quad (\text{because the prior series is geometric}). \end{aligned}$$

**Proof of Theorem 4.1** Following Lemma 4.1.1, the proof of The Collage Theorem is straightforward. Let  $L \in \mathcal{H}(\mathbf{X})$  be given, and suppose we find some IFS  $w_1, w_2, \dots, w_n$  with attractor  $A$  and contractivity factor  $s$  so that

$$h(L, \bigcup_{i=1}^n w_i(L)) \leq \epsilon.$$

In this scenario, we have a contraction mapping  $W(L) = \bigcup_{i=1}^n w_i(L)$  with fixed point  $A$  and contractivity factor  $s$ . So by lemma 4.1.1,

$$\begin{aligned} h(L, A) &\leq (1 - s)^{-1} h(L, W(L)) \\ \implies h(L, A) &\leq (1 - s)^{-1} \epsilon. \end{aligned}$$

■

How would we create the Figure 6 using The Collage Theorem?

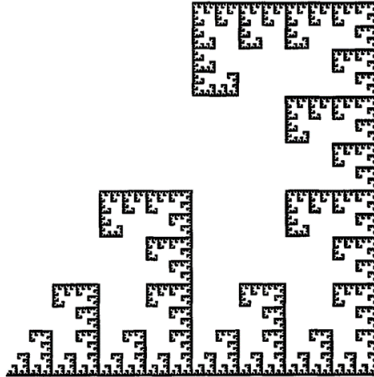


Figure 6: Because this set has self-similarity, we can use The Collage Theorem to find an IFS that approximates it [4].

After some inspection, we can see that the shape has self-similarity. In particular, if we partition the set into the bottom left, bottom right, and top right “squares,” we can see that each partition is similar to the whole set. Both the bottom left and bottom right partitions are equivalent to the whole set shrunk and translated, while the top right partition is the whole set shrunk, translated, and rotated. The Collage Theorem tells us that if the union of contraction mappings on a set is close to the original set itself, then the attractor of the IFS is close to the original set; therefore, we can represent the complicated shape with the following simple transformations.

$$\begin{aligned} w_{\text{bottom left}}(x, y) &= \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\ w_{\text{bottom right}}(x, y) &= \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 1/2 \\ 0 \end{bmatrix} \\ w_{\text{top right}}(x, y) &= \begin{bmatrix} 0 & -1/2 \\ 1/2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}. \end{aligned}$$

We can also create an IFS whose attractor approximates the leaf from Figure 5. The union of the images of the contraction mappings, depicted in Figure 7, are close to the original leaf; therefore, their attractor must also be close to the leaf.



Figure 7: The union of these contraction mappings is close to the original leaf, so their attractor is a good approximation of the leaf.

The contraction mappings depicted above can be written as follows.

$$\begin{aligned}
 w_{\text{red}}(x, y) &= \begin{bmatrix} 0.394 & -0.550 \\ 0.550 & 0.394 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -0.258 \\ 0.042 \end{bmatrix} \\
 w_{\text{purple}}(x, y) &= \begin{bmatrix} 0.579 & 0 \\ 0 & 0.579 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0.133 \\ 0.300 \end{bmatrix} \\
 w_{\text{blue}}(x, y) &= \begin{bmatrix} 0.261 & 0.661 \\ -0.661 & 0.261 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0.153 \\ -0.303 \end{bmatrix},
 \end{aligned}$$

The attractor of the IFS consisting of  $w_{\text{red}}$ ,  $w_{\text{purple}}$ ,  $w_{\text{blue}}$  is depicted in Figure 8.

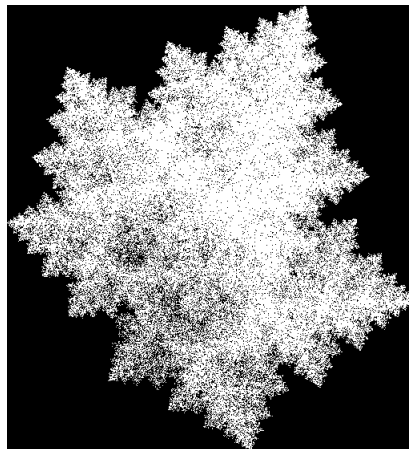


Figure 8: The image above is the attractor of the IFS described above. It approximates the leaf from Figure 5.

## 4.1 Continuous Changes in Attractors

Oftentimes, IFS's are used to generate images with computers. Complicated shapes like trees, leaves, clouds, or mountains can be represented with just a few functions. Contraction mappings on  $(\mathbb{R}^2, |\cdot|)$  are used to create shapes and might have some parameters in  $\mathbb{R}$  that vary. Consider the example of the leaf. What if we add some parameter  $p$  to the contraction mappings, as shown below?

$$\begin{aligned} w_{\text{red}}(x, y) &= \begin{bmatrix} 0.394 & -0.550 + p \\ 0.550 & 0.394 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -0.258 \\ 0.042 \end{bmatrix} \\ w_{\text{purple}}(x, y) &= \begin{bmatrix} 0.579 & 0 \\ 0 & 0.579 + p \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0.133 \\ 0.300 \end{bmatrix} \\ w_{\text{blue}}(x, y) &= \begin{bmatrix} 0.261 & 0.661 \\ -0.661 + p & 0.261 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0.153 \\ -0.303 \end{bmatrix}. \end{aligned}$$

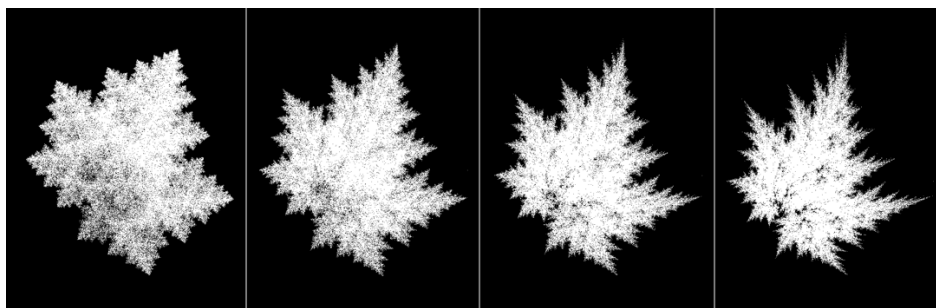


Figure 9: From left to right, we have the attractor of the IFS with  $r=0$ ,  $r=0.10$ ,  $0.15$ , and  $0.20$ .

In Figure 9, we can see that small changes in the parameter  $p$  lead to small changes in the attractor, making the leaf appear like it is blowing in the wind. Parameterizing contraction mappings in an IFS is useful for animating moving objects like clouds or trees.

In this section we show that under certain conditions, the attractor of an IFS varies continuously with the parameters of the contraction mappings. Particularly, we might have a function  $w : P \times \mathbf{X} \rightarrow \mathbf{X}$  that is a contraction mapping on the complete metric space  $(\mathbf{X}, d)$  and is continuous on the metric space  $(P, \rho)$ . For the remainder of this section, we denote a function  $w(p, x)$  on  $p \in P$  and  $x \in \mathbf{X}$  with  $w_p(x)$ . Our goal is to prove the following theorem [4].

**Theorem 4.2.** *The Continuous Dependence of Fractals on Parameters*

*Let  $w_1, w_2, \dots, w_n$  be an IFS on the complete metric space  $(\mathbf{X}, d)$  with contractivity factor  $s$ . Suppose that each  $w_i$  depends on a parameter in  $(P, \rho)$ . Moreover, suppose that there*

exists  $k > 0$  such that

$$d((w_i)_p(x), (w_i)_q(x)) \leq k \cdot \rho(p, q)$$

for all  $x \in \mathbf{X}$  and for all  $p, q \in P$ . Then the attractor of the IFS  $A_p \in \mathcal{H}(\mathbf{X})$  depends continuously on the parameter  $p$  with respect to the Hausdorff metric.

First, consider the following lemma.

**Lemma 4.2.1.** *Suppose  $(P, \rho)$  is a metric space and  $(\mathbf{X}, d)$  is a complete metric space. Furthermore, suppose  $w : P \times \mathbf{X} \rightarrow \mathbf{X}$  is a contraction mapping on  $(\mathbf{X}, d)$  and is continuous on  $(P, \rho)$ . Then the fixed point  $x'_p : P \rightarrow \mathbf{X}$  is continuous.*

**Proof of Lemma 4.2.1**

Suppose  $(P, \rho)$  is a metric space and  $(\mathbf{X}, d)$  is a complete metric space. Furthermore, suppose  $w : P \times \mathbf{X} \rightarrow \mathbf{X}$  is a contraction mapping on  $(\mathbf{X}, d)$  and is continuous on  $(P, \rho)$ . Let the parameters  $p, q \in P$  be given. Recall from lemma 4.1.1 that for any contraction mapping, with contractivity factor  $s$  and fixed point  $x'$ ,

$$d(x, x') \leq (1 - s)^{-1} \cdot d(x, w(x)).$$

Let  $p, q \in P$  be given so that  $w_p$  has fixed point  $x'_p$  and contractivity factor  $s_p$ . Similarly, suppose  $w_q$  has fixed point  $x'_q$  and contractivity factor  $s_q$ . We can apply lemma 4.1.1 taking  $w_q$  as our contraction mapping and  $x'_p$  to be our arbitrary point  $x$ :

$$d(x'_p, x'_q) \leq (1 - s_q)^{-1} \cdot d(x'_p, w_q(x'_p)) \quad (1)$$

$$\leq (1 - s_q)^{-1} \cdot d(w_p(x'_p), w_q(x'_p)) \quad (\text{because } x'_p \text{ is a fixed point}). \quad (2)$$

Now, let  $\epsilon > 0$  be given. Because  $w$  is continuous on  $P$ , we can take a  $\delta > 0$  so that  $\rho(p, q) < \delta$  implies  $d(w_p(x), w_q(x)) < \epsilon \cdot (1 - s_q)$  for all  $x \in \mathbf{X}$ . Then combining this inequality with what we have for (1),(2)

$$\begin{aligned} d(x'_p, x'_q) &\leq (1 - s_q)^{-1} \cdot d(w_p(x'_p), w_q(x'_p)) \\ &< (1 - s)^{-1} \cdot \epsilon \cdot (1 - s_q) \\ &= \epsilon. \end{aligned}$$

In conclusion, we have shown that for all  $\epsilon > 0$  there exists some  $\delta > 0$  so that

$$\rho(p, q) < \delta \implies d(x'_p, x'_q) < \epsilon,$$

so the fixed point  $x'_p : P \rightarrow \mathbf{X}$  is continuous by definition.  $\square$

We want to eventually show that for an IFS  $w_1, w_2, \dots, w_n$  on a complete metric space  $(\mathbf{X}, d)$  with continuous dependence on some parameter  $p \in (P, \rho)$ , the contraction mapping  $W : \mathcal{H}(\mathbf{X}) \rightarrow \mathcal{H}(\mathbf{X})$  given by  $W(B) = \bigcup_{i=1}^n w_i(B)$  has an attractor  $A_p : P \rightarrow \mathcal{H}(\mathbf{X})$  that is continuous. Although we have already shown that  $W$  is a

contraction mapping, we don't know if it depends continuously on  $p$ ; therefore, lemma 4.1.1 is not enough to give us theorem 4.2. We need another condition on the contraction mappings  $w_1, w_2, \dots, w_n$  such that for all  $\epsilon > 0$  and for all  $B \in \mathcal{H}(\mathbf{X})$ ,

$$\rho(p, q) < \delta \implies h(w_p(B), w_q(B)) < \epsilon.$$

One condition that works is called *Lipshitz continuity*.

**Definition 8.** *Lipshitz Continuity*

A function  $f : (P, \rho) \rightarrow (\mathbf{X}, d)$  is *Lipshitz continuous* if there exists some  $k > 0$  so that

$$\rho(f(x), f(y)) \leq k \cdot d(x, y)$$

for all  $x, y \in P$ .

Now, we can prove the theorem

**Proof of Theorem 4.2**

Suppose that for every  $p \in P$ ,  $(w_i)_p$  is a contraction mapping on the complete metric space  $(\mathbf{X}, d)$ . Furthermore, suppose that  $w_i : P \times \mathbf{X} \rightarrow \mathbf{X}$  is Lipschitz continuous on  $P$ . That is, there exists some fixed  $k > 0$  such that

$$d((w_i)_p(x), (w_i)_q(x)) \leq k \cdot \rho(p, q)$$

for all  $p, q \in P$ . We want to show that for all  $B \in \mathcal{H}(\mathbf{X})$ ,  $h((w_i)_p(B), (w_i)_q(B)) \leq k \cdot \rho(p, q)$ . Let  $p, q \in P$  be given and consider some arbitrary  $x \in (w_i)_p(B)$ . By definition of an image, there exists some  $\tilde{x} \in B$  such that  $(w_i)_p(\tilde{x}) = x$ . because  $w_i$  is Lipschitz continuous on  $P$ ,

$$d((w_i)_p(\tilde{x}), (w_i)_q(\tilde{x})) = d(x, (w_i)_q(\tilde{x})) \leq k \cdot \rho(p, q)$$

for some fixed  $k > 0$ . This implies that

$$\min_{y \in (w_i)_q(B)} d(x, y) \leq k \cdot \rho(p, q)$$

because  $\min_{y \in (w_i)_q(B)} d(x, y) \leq d(x, (w_i)_q(\tilde{x}))$ . Moreover, since this inequality holds for every  $x \in (w_i)_p(B)$  we have that

$$\begin{aligned} \max\{\min\{d(x, y) : y \in (w_i)_q(B)\} : x \in (w_i)_p(B)\} &\leq k \cdot \rho(p, q) \\ \implies d((w_i)_p(B), (w_i)_q(B)) &\leq k \cdot \rho(p, q). \end{aligned}$$

By the same argument, we can show that  $d((w_i)_q(B), (w_i)_p(B)) \leq k \cdot \rho(p, q)$ ; therefore,

$$h((w_i)_q(B), (w_i)_p(B)) \leq k \cdot \rho(p, q). \quad (1)$$

Intuitively, we have shown that a small change in the parameter  $p$  produces a small change in the image of the map  $(w_i)_p(B)$ . Suppose we have finitely many maps  $(w_1)_p, (w_2)_p, \dots, (w_n)_p$  that all vary with parameter  $p$ . Suppose  $W_p(B) = \bigcup_{i=1}^n (w_i)_p(B)$ . The union of the images cannot vary more than the individual images vary, so if  $k = \max\{k_1, k_2, \dots, k_n\}$ , the max of the Lipschitz constants for all the functions, then for all  $B \in \mathcal{H}(\mathbf{x})$

$$h(W_p(B), W_q(B)) \leq k \cdot \rho(p, q).$$

Now, we have that  $W_p$  is a contraction mapping and  $W_p$  is continuous on  $P$ , so we can apply lemma 4.2.1; the attractor  $A_p$  of the IFS is a continuous function on  $P$ . ■

Recall our example of the leaf in the beginning of this section. Since each of the contraction mappings is Lipschitz continuous on the parameter space  $(\mathbb{R}^2, |\cdot|)$ , the attractor is continuous on  $(\mathbb{R}^2, |\cdot|)$ .

## 5 Fractal Image Compression

The Collage theorem has found its main application in image compression. We can take advantage of self-similarity within an image to represent it with a small number of functions rather than many pixels. With what we have learned so far, we know that if we can find a series of contraction mappings on an image whose union is close to the original image, the attractor of the IFS is also close to the original image. It's usually not the case, however, that an entire image is self-similar to parts of the image. Instead, we can partition an image and find parts that are similar to other parts. For instance, in the image below, edges are similar to other edges, grey areas are similar to other grey areas, and portions with changing color gradients are similar to other areas with changing gradients. How can we automate the process of finding an IFS that matches similar parts? I will describe the process outlined by Jacquin [5].



Figure 10: The boxed edges in the image are self-similar [6].

Suppose we have a metric space of grayscale images  $(\mathcal{M}, d)$ .  $d$  can be any metric that measures the difference between images. For example, suppose  $d$  is the sum of the squared differences between pixel values in two images. Consider some image  $\mu \in \mathcal{M}$ . We first want to create two partitions of  $\mu$  into squares:  $\{D_j\}$  and  $\{R_i\}$  where the squares in  $\{R_i\}$  are smaller. For example, suppose that the squares  $\{D_j\}$  have size  $B \times B$ , and the squares  $\{R_i\}$  have size  $B/2 \times B/2$ . For each  $R_i$  we want to find contraction mappings  $\tau_i : D_j \rightarrow R_i$  that minimize  $d(\tau_i(D_j), R_i)$ . We can imagine that each contraction mapping  $\tau_i$  performs two functions:  $\tau_i$  maps a domain block  $D_j$  to  $R_i$ , and it distorts the domain block  $D_j$ . Therefore, we can denote  $\tau_i$  as follows.

$$\tau_i = \psi_i \circ \phi_i$$

where  $\phi_i$  is the *geometric part* that maps a domain block to the range block, and  $\psi_i$  is the *massic part* that distorts the domain block. To find the optimal contraction mapping  $\tau_i$ , for each range block  $R_i$ , we need to search for an optimal pair  $(D_i, \psi_i)$  that minimizes the distortion  $d$ . Note, that finding an optimal domain block  $D_i$  is equivalent to finding the optimal geometric part  $\phi_i$ . There are infinitely many possible pairs  $(D_i, \psi_i)$ , so we need to limit the pool in a way that allows us to search through the entire pool and find the optimal pair.

First, we group the domain and range blocks into *shade blocks*, *edge blocks*, and *mid-range blocks* using the algorithms proposed by Ramamurthi and Gersho [7]. Shade blocks are “smooth” and have no significant gradient change; edge blocks have a strong contrast in color, like a line or edge in an image; and mid-range blocks have some gradient change or some patterned texture. Shade blocks are excluded from the search pool of possible  $D_i$ ’s; if  $R_i$  is a shade block, we simply approximate it with a uniform gray block with gray level equal to the average gray level of  $R_i$ . If  $R_i$  is an edge block or mid-range block we want to restrict the possible massic parts  $\psi_i$ —again reducing an infinite search to a finite one. For mid-range blocks,  $\tau_i$  can take the form

$$\tau_i(D_j) = \alpha_i(\phi(D_j)) + \Delta g_i$$

where  $\alpha_i$  is an algorithm that matches the contrast of  $D_i$  and  $R_i$ , and  $\Delta g_i$  is computed, so the average gray levels are the same in  $D_i$  and  $R_i$ . For more information on image processing and algorithms such as  $\alpha$  see Section 3.3 of *Digital Image Processing* on histogram processing [8]. Similarly, if  $R_i$  is an edge block, then we also match the contrast, with the algorithm  $\alpha$  and match the average gray scales by adding  $\Delta g_i$ , but we should also compare different isometries (i.e. transformations that preserve distances in the image). There are eight different isometries that we should consider: the identity transformation; reflections across the horizontal, vertical, and both diagonal axes; and rotations of  $90^\circ$ ,  $180^\circ$ , and  $270^\circ$ . Denote these isometries as  $l_{n_i}$ . Then our possible contraction mappings are given by

$$\tau_i(D_j) = l_{n_i}(\alpha(\phi(D_j)) + \Delta g_i).$$

Now, for each  $R_i$ , we have reduced our pool of possible domain blocks and massive transformations  $(D_i, \psi_i)$  to finitely many options. We should search through each to find the pair that minimizes the distortion  $d(\tau_i(D_i), R_i)$ . Doing this for each  $R_i$  minimizes the distortion

$$d\left(\bigcup \tau_i(D_i), \bigcup R_i\right) = d\left(\bigcup \tau_i(D_i), \mu\right) \quad (1)$$

where we are taking the union over all range blocks  $R_i$  in the image, and each  $(D_i, \tau_i)$  is the corresponding domain block and transformation. Notice that the number of functions  $\tau_i$  that we have depends on the number of range blocks  $R_i$ . If we combine the operations of each  $\tau_i$  we get a contraction mapping  $\tau : (\mathcal{M}, d) \rightarrow (\mathcal{M}, d)$ . Moreover, from (1) we have minimized

$$d(\tau(\mu), \mu).$$

If our algorithm to minimize the distortion is “good,” we may have that

$$d(\tau(\mu), \mu) \leq \epsilon$$

for some small  $\epsilon > 0$ . Then by theorem 4.1 (*The Collage Theorem*) we have that

$$d(\mu, A) \leq \epsilon/(1 - s) \quad (2)$$

where  $s$  is the contractivity factor, and  $A$  is the fixed point of  $\tau$ . In other words, the fixed point of the contraction mapping  $\tau$  is close to our original image  $\mu$ .

We have reduced the information in many pixels into fewer simple functions  $\tau_i$ . Using these functions, we can recreate an image close to our original image  $\mu$ . In the example below from Jacquin [5], the image on the left is recreated on the right by iterating  $\tau$  on a uniform gray image three times.



Figure 11: The image “Lena” on the left is recreated on the right by iterating  $\tau$  on the gray image three times.

The information in image was compressed by a factor of 12.5, and the reproduced image is close to the original after only three iterations. If we had iterated  $\tau$  several more times, we would anticipate that the reproduced image would become closer and closer to the original.

## 6 Conclusion

Using The Collage Theorem, we can often represent complex shapes with only a few simple functions. The one requirement is that the shapes must exhibit some self-similarity. Fortunately, many shapes in nature like leaves, trees, and clouds have self-similarity. We have seen how we can use The Collage Theorem to easily find an IFS to represent these shapes. Moreover, we have seen how we can parameterize an IFS to create continuous changes in its attractor, allowing us to simulate movement like blowing in the wind. Lastly, we learned that we can use The Collage Theorem to take advantage of self-similarity in images and compress them. Although we have focused on applications in  $(\mathbb{R}^2, |\cdot|)$ , everything has been proven rigorously for metric spaces in general. The Collage Theorem can be used to uncover patterns and compress information in many spaces.

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